

Glass Is an Unclosed Layer: Kovacs Memory as Predictive-Quotient Residue

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Abstract

Sixty years of searching have produced no universally accepted static order parameter for glass, and the Kovacs crossover experiment of 1963 illustrates the difficulty: at the crossing moment the glass equals its equilibrium in the monitored quantity and then departs from it, so that quantity alone does not fix what happens next. We turn that observation into theorems on a small, exactly computable class of kinetically constrained models (East, Fredrickson–Andersen, and a Bouchaud-type trap family), where kernels, stationary measures, witnesses, and zero controls are ratios of integers and the remaining certificate evidence is deterministic float64, labeled as such in each artifact. The question we make precise is whether a declared readout forms a *closed* description: whether its present value determines its own future statistics. On the declared scope we show: the energy readout of an aging East chain is far from closed, its closure deficit exceeding the constrained-equilibrium baseline by a certified margin of $3/2$ (achieved ≈ 1.64), and that baseline is itself provably nonzero, because closure is a property of the kinetic constraint rather than of equilibrium (GT1); the Kovacs protocol yields an exact pair of preparation histories with identical present mean energy and different futures (GT2); all 512 yes/no functions of the declared energy readout agree on such a pair while a finer readout of the same microstate separates it, so no static order parameter built from the energy readout separates, on that class, what the pair’s futures separate (GT3); five constructed look-alike regimes are correctly rejected by the same classifier that accepts the Kovacs candidate (GT4); one added scalar coordinate, a fictive-temperature-like index among others, resolves the pair, which demonstrates the repair mechanism without proving that one scalar is insufficient in general (GT5); the description of histories by future-equivalence strictly extends the declared energy-based description, certified for the audited shell (GT6); and aging constants frozen on East and transferred to FA under a pre-registered no-fitting protocol scored 8 passes and 14 one-sided failures, with the trap family unobservable for a reason we prove: the declared trap weight family’s stationary distribution is exactly temperature-invariant (GT7). A vendored Lean core machine-checks the abstract witness and refinement theorems; attempting to encode the numeric witness produced a machine-checked impossibility theorem that locates that core’s expressiveness boundary. Every claim is hash-pinned to its artifacts, and the full ledger of what is *not* claimed is printed unabridged.

Keywords. glass transition; Kovacs effect; aging and memory; kinetically constrained models; East model; fictive temperature; closure deficit; lumpability; predictive quotient; Six Birds Theory; emergence calculus; exact finite certificates; pre-registered prediction; Lean.

1 Introduction

1.1 The problem

Anderson called the nature of glass and the glass transition probably the deepest and most interesting unsolved problem in solid-state theory [3]. One face of that problem is sixty years old and unusually concrete: no universally accepted static order parameter for glass has emerged. A glass and a liquid can agree in density, energy, and every routinely monitored one-time observable, yet behave differently [26]; conventional low-order static correlations show no clear signature of the transition, and the candidate structural measures that do grow remain contested [8, 19, 68, 61]. A second face has sat in the laboratory record since 1963: Kovacs’s crossover experiment [40]. Equilibrate a polymer glass hot, quench it, age it, then jump the temperature up and watch its volume relax. At some moment the volume *equals* its new equilibrium value, and does not stay there: it overshoots through a non-monotone hump before returning. At the crossing moment the sample is indistinguishable from equilibrium in the monitored observable and distinguishable from it in its future, a point made explicit at the level of configurations in simulation [46]. The standard partial fix, the Tool–Narayanaswamy–Moynihan fictive temperature [69, 49, 47], adds one memory variable; richer protocols expose the limits of any single such variable and have motivated multiparameter models [58, 41, 33, 43]. Memory effects of this kind, in which a material’s response depends on its history in a way no monitored one-time observable records, are now studied across many classes of matter [37].

1.2 The question, made precise

Both faces are the same question. Take a declared readout of the system (here, the energy of a spin chain; in the laboratory, the volume) and ask: *is that readout a closed description of the dynamics?* A readout is closed over a horizon τ if its present value determines the statistics of its value τ steps later, so that a law can be written in the readout’s vocabulary without loss, the informational closure of [10, 54]. In Markov-chain language the readout is closed at horizon τ exactly when the τ -step readout transition probabilities agree across every populated microstate with the same readout, which for a fully supported distribution is lumpability of the readout [17, 38]. In this paper a static order-parameter candidate is taken in one operational sense only: a function of the declared readout that separates two preparations whose futures differ. Any function of the readout that fails to separate them (a constant does so trivially) cannot carry the information the futures carry, and that is the sense in which GT3 below forecloses the search: not that no coarsening of the energy is closed, but that none of them tells the exhibited pair apart. That is narrower than the field’s usage. The Kovacs hump is a direct demonstration that the monitored volume is not closed.

Once the question is put this way, three further questions become computable on a finite model. How far from closed is the readout, in nats? Can one exhibit a concrete certificate of the failure, that is, two preparation histories that agree on the readout now and differ in their futures? And is there *any* function of the readout that recovers the missing predictive distinction, or must it be carried by a genuinely new coordinate, and at what cost?

1.3 What this paper does

We answer those questions on a class of models small enough that the certificates are exact ratios of integers wherever the artifact declares them so, and deterministic float64 elsewhere: kinetically constrained models [59, 31], with the East model [35] as the substrate on which everything is built and calibrated, and the Fredrickson–Andersen model [29] and a Bouchaud-type trap family [12, 44]

as out-of-sample transfer targets. At the system sizes used ($N \leq 10$ spins for certificates), transition kernels, stationary measures, witness pairs, and zero controls are ratios of integers; the closure-deficit grid of GT1 is deterministic float64 and is labeled as such in its artifact, with its margin far from the rounding scale. The price is that every result is a statement about a declared finite class of models, sizes, observables, and protocols, and we state that scope in every claim.

The bookkeeping that keeps those scopes honest is borrowed from a certificate-based closure calculus, Six Birds Theory [77, 70, 71, 72, 73, 74, 75, 76]. The reader does not need that corpus to follow this paper: §2 defines every object we use in ordinary probabilistic terms, and the framework’s own labels appear only where a claim is filed in its vocabulary, always with a plain gloss. Readers who want the formal anchors will find each claim’s corpus reference in the claims ledger shipped with the paper.

The results, each at its declared scope (§4–§8), read as follows.

- **GT1, the energy readout is not closed.** For an aging East chain, the closure deficit of the energy readout (the conditional mutual information between the microstate now and the energy later, given the energy now) exceeds the same quantity at constrained equilibrium by a certified factor of at least $3/2$. A correction to our own design expectations came with it: the equilibrium value is provably *not* zero, because closure depends on where the mobile excitations sit, not on whether the chain is stationary.
- **GT2, the Kovacs witness.** The Kovacs protocol on East yields an exact pair of preparation histories with the same present mean energy and different future mean energies under the declared continuation.
- **GT3, the order-parameter search is foreclosed on the declared class.** Every one of the 512 yes/no functions definable from the declared energy readout agrees on a witness pair that a finer readout of the same microstate (the occupation of the site next to the wall) separates. On that class, no static order parameter built from the energy readout can do the job; whatever does the job must carry information the energy readout does not.
- **GT4, the kill-test panel.** The classifier that accepts the Kovacs memory candidate correctly rejects five constructed look-alike regimes, schematic fixtures for a protocol artifact, dissipation, an explicitly latent variable, an equilibrium control, and an incomplete continuation catalog.
- **GT5, the cost of repair.** One added scalar coordinate resolves the witness pair, and a fictive-temperature-like index (the index of the grid point whose equilibrium mean energy is closest to each microstate’s energy, averaged over the history) is one of three that do. The certificate shows the repair mechanism on this pair; it does not prove that one scalar is insufficient in general.
- **GT6, the memory layer is strictly richer.** The description of histories by future-equivalence strictly extends the declared energy-based description, certified for the audited shell (the frozen set of eligibility conditions, §2) by an explicit gate table and ablation panel.
- **GT7, aging constants transfer only partly.** Six aging readouts were frozen on East, predictions for FA and the trap family were pre-registered, and the targets were then run and scored mechanically: 8 of 22 scored predictions passed, 14 failed on the same side, and the trap family produced no Kovacs hump at all, for a reason we prove.

Two further results come from the formalization effort (§9). A vendored core of the framework’s theorems is machine-checked in the Lean proof assistant, and the attempt to encode the numeric

Kovacs witness in that core produced a machine-checked impossibility theorem instead, which locates exactly which certificates can live in that core and why.

1.4 How to read the claims

Four contributions, in the order we would defend them. First, the **discipline**: every claim quantifies over a declared finite class, is pinned by content hash to the artifact that supports it, is accompanied by controls, and is filed together with what it does *not* assert; honest negatives are results, and the discipline forced several of our own claims to shrink. Second, the **result complex** itself: seven claims, four bridges, and six honest negatives, each hash-pinned to its artifacts. The bracketed labels throughout, such as [GT1], are machine-checked citations into the claims ledger, not decoration. Third, the **freeze-and-transfer protocol with its actual scoreboard**: the failures are results, with a clean one-sided pattern and a proven structural explanation for the empty column. Fourth, the **Lean-checked core and the scope-boundary finding**.

What this paper is not: it proves nothing about all glasses, the continuum, or the ideal-glass transition; it makes no irreversibility claims; and its field-facing sentence, “no static order parameter,” is licensed only over the declared class it is proved on. The full list of nonclaims is printed, unabridged, in §11.

2 Definitions and Methods

2.1 The objects, in plain terms

Everything in this paper is built from a finite-state Markov chain and a choice of what to observe. Five objects carry all seven results; we define each in ordinary terms, with the Kovacs experiment as the running example, and give its source in the Six Birds Theory (SBT) corpus, which we cite rather than re-derive [77, 70, 71, 72, 73, 74, 75, 76].

Lens. The observable panel: a map f from the microstate space Z to a finite set of readout values.

Our main lens is the energy readout L_{energy} , the number of up spins; the “thermodynamic lens” Π_{std} is this readout together with its declared panel. In the laboratory the lens is the dilatometer. (SBT: a finite theory package, F-I [70], Def. 1.)

Closure and the closure deficit. A lens is *closed* at horizon τ , for a given law and a given distribution over microstates, if the readout now determines the distribution of the readout τ steps later, with nothing further gained from the microstate. The *closure deficit*

$$\text{CD}_\tau(\Pi) = I(X_t; Y_{t+\tau} | Y_t)$$

is the conditional mutual information, in nats [23], between the microstate X_t and the future readout $Y_{t+\tau}$ given the present readout Y_t , under the declared law and state distribution. It is zero exactly when the τ -step readout transition probabilities are the same from every *positive-mass* microstate with the same readout; when the declared distribution has full support this is τ -lumpability of the (kernel, lens) pair itself [17, 38, 15], and a zero-mass region is never audited (the distribution-dependent situation studied as weak lumpability [62]; a point-mass start on East $N = 2$ has $\text{CD} = 0$ although the kernel is not lumpable, a regression-tested control). Lumpability at one horizon does not by itself imply it at others, so the τ ladder is declared. When positive it is an exact lower bound on what any autonomous law at that horizon, written in the lens’s vocabulary, must discard. This is the informational-closure criterion of [10, 54]; a

readout that is not closed obeys a law with a memory kernel, in the projection-operator sense of [81, 45]. (SBT: F-III [72] T8; [75] Def. 3.1.)

Packaging map and idempotence defect. A second, coarser diagnostic. Evolve the chain τ steps, push the distribution through the lens, and lift it back to a microstate distribution by a declared rule (the “prototype” for each readout value): $E_{\tau,f}(\mu) = U_f(Q_f(\mu P^\tau))$. The *idempotence defect* $\delta_{\tau,f}$ is the largest total-variation distance, over starting microstates, between applying this map twice and applying it once. It is a saturation diagnostic and nothing more: a lumpable lens can still have $\delta_{\tau,f} > 0$ (the unconstrained control at $N = 2$, $\epsilon = 1/2$, $\tau = 1$ has exact defect $2/9$, and this is a regression test), so the defect is reported in GT1 only as a ratio between the aging protocol and its equilibrium baseline, never as a closure criterion. (SBT: F-I Def. 9.)

Preparation histories, quotients, and witnesses. A *preparation history* is a protocol word (“equilibrate at ϵ_1 , quench to ϵ_2 , age t_w , jump to ϵ_3 ”) together with the state distribution it produces. Over a declared finite catalog of histories and continuation protocols, the *current quotient* Q identifies histories with equal present lens readouts, and the *predictive quotient* M identifies histories with equal future readout statistics under every declared continuation (the causal-state construction of computational mechanics [24, 63, 64], restricted to a declared finite catalog). Under the framework’s definitions M refines Q , so there is always a comparison map $\pi : M \rightarrow Q$. A *witness*, or split pair, is two histories h, h' that Q identifies and M separates: same now, different later. Witnesses exist exactly when π merges distinct M -classes (“strictly refining”). The Kovacs sample at its crossing moment and an equilibrated sample at the same volume are the laboratory picture of a witness pair. (SBT: [74], Defs. 2.2–2.4 and Thm. 3.5; machine-checked as `strictRefinement_iff_nonempty_predictiveWitness`, §9.)

Memory coordinates and buyback. A witness pair can be resolved by adding coordinates to the lens. The *buyback curve* records, for each budget of added scalars, which declared candidate coordinates resolve the pair; each such addition is a recorded, selected, non-unique choice (a “lift” in SBT terms). Fictive temperature is one such candidate. (SBT: F-II `prop:forgetting-lifts`.)

Four more pieces of vocabulary recur. A *lens-definable predicate* is a yes/no function of the readout value; a static order parameter built from the lens is one of these. A *shell* is a frozen set of eligibility conditions under which a certificate is claimed: for GT6, the model and N , the declared ϵ grid, the declared τ windows per protocol leg, a non-equilibration criterion at the measurement time, and the presence of a Kovacs hump, each checked by a membership predicate with an exit counter that must read zero. A *strict extension* (SBT: a strict promotion) is the statement that the predictive description M carries information the declared present description Q cannot, certified by an explicit checklist (a “gate table”) and by ablations (a “knockout panel”). And the *six roles* P1–P6 are the framework’s fixed list of ways a description can fail to close; the only ones this paper files are P4 (“the future depends on more than the present readout”) and P5 (“the packaging of the present readout is not a closed projection”). Where a claim is filed in that vocabulary, we say so and gloss it.

Claim grades and nonclaims. Every result carries exactly one grade, taken from the framework’s fidelity scale (F-II [71], §9.3): *theorem-grade* (*audited class*), proved or audited unconditionally within the covered finite scope, at the artifact’s declared evidence type (exact rational or deterministic float64); *certificate-grade* (*shell-local*), the standard of paper [C] [76], valid on the frozen shell with no broader-class transfer; *run-only readout*, a measured number with no model fitted to it;

recognition-grade, a demonstration rather than a theorem; or *control*. Every artifact carries a machine-readable envelope (`claim_id`, `claim_grade`, `quantifier_domain`, `nonclaims`) validated against a schema in the reproducibility package. Nonclaims are not boilerplate: the full nonclaims ledger is printed in §11, and several are load-bearing (for example GT6’s verdict basis, §7).

2.2 Exact arithmetic and the freeze discipline

Exact rational arithmetic (Python `fractions.Fraction`) carries the kernels, stationary measures, witness gaps, the GT3 predicate sweep, the trap-invariance lemma, the GT1 idempotence-defect ratio (289/121), and the lumpable zero control (whose CD is exactly 0 because exact lumpability is verified symbolically). Not everything theorem-graded is rational, and each artifact says which it is: GT1’s CD grid is deterministic float64 evidence (`evidence_type: float64_deterministic` in the artifact itself), and its declared margin 3/2 is checked in float64. The claim’s grade rests on the audited-class quantification plus the artifact’s declared evidence type, not on a pretended exactness. Float64 also appears in explicitly labeled diagnostics (spectral gaps, windowed CD diagnostics, GT7 Monte Carlo spot checks, the GT6 pressure closure). Controls accompany every certificate, including the constructed negative controls of the triage panel (§6).

Pre-registration (GT7, §8) and the paper’s own evidence base are frozen by sha256 content hashes over canonical JSON. Each freeze artifact carries a `content_hash` and a `freeze_note`, and every cross-artifact citation pins the target’s hash, so regenerating any artifact invalidates, detectably, every citation of it. A repository-wide test suite enforces all such bindings, and the paper’s own claim citations (the bracketed row labels throughout, e.g. [GT1]) are pinned the same way: each carries a hidden artifact hash checked against the claims ledger and the current artifact bytes at build time. Version-control history is deliberately not the freeze mechanism; the content hashes are the proof object.

2.3 What the reader should check

The claims ledger (`paper/claims_ledger.json`) is the paper’s load-bearing skeleton: one row per claim with final wording, grade, quantifier domain, anchors into the SBT corpus, artifact hashes, and nonclaims. Every results section below is a transcription of its ledger rows into prose; any discrepancy between prose and ledger is a bug in the prose. A reader with the repository can re-run every certificate: the test suite, the Lean build, and the three checkers (claims ledger, GT6 filing, empirical bridges) all run from a clean checkout.

3 Models and Pipeline

3.1 Model families

Four model families populate the audited class; all are finite-state discrete-time Markov chains with exactly rational kernel entries. Three are kinetically constrained models, whose glassy phenomenology comes from constraints on the dynamics rather than from their trivial product statics [30, 20]. Temperature never enters as a free real parameter. Instead we declare a finite rational grid in the Boltzmann activation variable $\epsilon \in \{1/10, 1/5, 3/10, 2/5, 1/2, 3/5, 7/10\}$, where ϵ plays the role of $e^{-\beta}$; the nominal temperature $T = -1/\ln \epsilon$ is an annotation for the reader and never a computational input. The up-spin equilibrium concentration $c = \epsilon/(1 + \epsilon)$ is exactly rational. A temperature protocol is a finite word over this grid, and the position in the protocol is made part of the state, so

that the chain stays autonomous and time-homogeneous even while the temperature schedule runs (the clock-audit discipline of F-I [70]).

East model (primary substrate). N binary spins with a frozen up-spin wall at site 0; spin i may flip only if its left neighbor is up ($C_i(n) = n_{i-1}$) [35], the one-dimensional realization of hierarchically constrained dynamics [51]. One step of the random-scan heat-bath (Glauber) kernel K_ϵ [32]: pick a site uniformly; if unconstrained, resample it from the Bernoulli(ϵ) marginal. Because the constraint does not involve the flipping spin itself, K_ϵ is reversible with respect to the product measure $\pi_\epsilon(n) \propto \epsilon^{E(n)}$ with $E(n) = \sum_i n_i$, verified programmatically as exact Fraction identities for every ϵ on the grid. One point matters later: K_ϵ and $K_{\epsilon'}$ are reversible with respect to *different* measures. That is exactly the hypothesis that the published protocol-trap theorem needs and a temperature protocol violates, which is what the GB1 demo (§6) addresses.

Fredrickson–Andersen (FA-1f). Identical structure with the symmetric constraint $C_i(n) = \max(n_{i-1}, n_{i+1})$ [29]. FA is the transfer target for GT7: nothing about FA is used in fitting or freezing; it is run only after predictions are committed (§8).

Unconstrained control. The same heat-bath chain with $C_i \equiv 1$. Here the energy readout is lumpable at every horizon (each site resamples independently, so the number of up spins is itself a Markov chain), and the closure deficit is exactly 0 as a rational number. This is the zero control for GT1 (§4).

Bouchaud-type trap model (second transfer target). M traps with declared rational escape weights w_k ; from trap k , jump to a uniformly random trap with probability proportional to w_k , else stay [28, 12, 44]. Temperature dependence is a declared finite family of rational weight vectors, one per grid ϵ (exponentiating weights would break rationality); the physical Arrhenius relation is again an annotation, not an input. The stationary measure is $\pi_k \propto 1/w_k$, verified exactly.

3.2 A proven limitation of the declared trap family

[FINDING.trap_epsilon_invariance] The declared trap weight family multiplies every trap’s base weight by a common per- ϵ scalar. Since $\pi_k \propto 1/w_k$ depends only on weight *ratios*, that scalar cancels exactly: the trap stationary distribution is ϵ -invariant (proved in exact Fraction arithmetic; regression-tested). The declared temperature action therefore governs relaxation *rate* only, never the relaxation *target*. This is known in substance in the trap-model literature [12, 44, 57, 6, 27, 9]: the standard convention is a per-trap Arrhenius factor $w_k(T) \propto e^{-E_k/T}$, under which different traps rescale by different temperature-dependent factors and the equilibrium genuinely shifts, which is exactly the mechanism used to produce Kovacs memory in trap models [27, 9]. Our multiplicative-scalar choice was made deliberately to keep every computation exactly rational; the invariance is its elementary, proven consequence. It is filed here as a design caveat because it has a concrete downstream cost, paid in §8: a Kovacs protocol cannot start the trap family below its post-jump equilibrium, so the trap column of the transfer scoreboard is structurally unobservable.

Nonclaim: Not a novelty claim: the cancellation is standard trap-model structure once stated; the contribution is stating, proving, and paying for it within the declared family. It says nothing about trap dynamics or aging trajectories; only the stationary distribution is ϵ -invariant.

3.3 Pipeline

The exact-finite pipeline ports paper [A]’s package specification [74] to thermal semantics: rational history distributions over a declared preparation catalog, rational row-stochastic continuation kernels for each protocol word, and two-time observables as future-signature entries. Parity with [A]’s benchmark family (a six-regime control suite, re-instantiated with thermal semantics) is filed as a control artifact and consumed by GT4 (§6). Spectral gaps of K_ϵ (float64 diagnostics, never certificate content) calibrate the τ windows and aging times used throughout (Fig. 1); their rapid closing as ϵ decreases is the East model’s super-Arrhenius relaxation, established numerically and rigorously in [66, 2, 18].

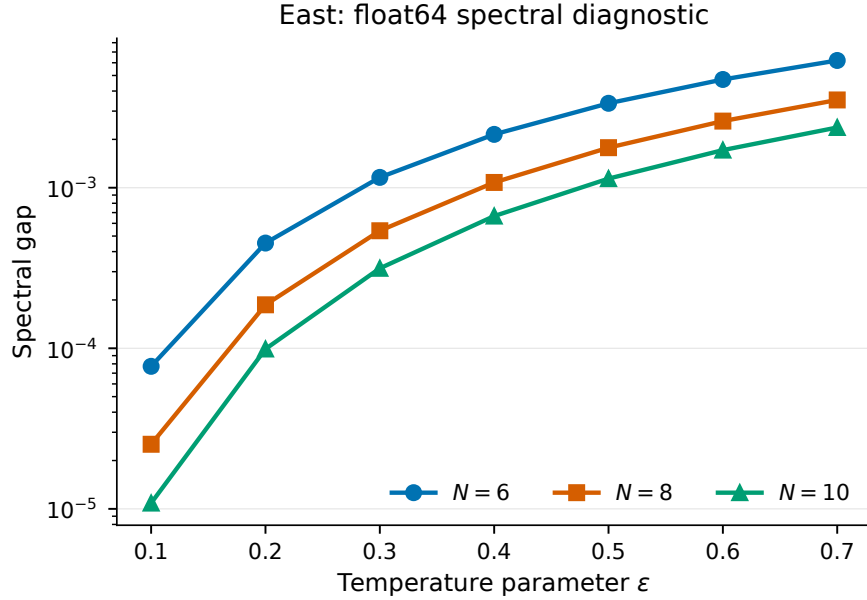


Figure 1. East-model spectral gaps across the declared (N, ϵ) grid (float64 diagnostic used only for window calibration).

4 GT1: The Energy Readout Is Not Closed

4.1 Claim

In plain terms: for an aging East chain, knowing the microstate tells you substantially more about the energy a few steps ahead than knowing the present energy does, and it does so by a certified factor more than at constrained equilibrium. The packaging defect of §2 is reported alongside as a second ratio. Precisely:

[GT1] On the declared scope (East model, $N = 8$, lens L_{energy} , τ ladder $[1, 2, 4, 8]$, quench cell $\epsilon : 7/10 \rightarrow 1/10$ with waiting time $t_w = 20$), the closure deficit CD_τ of the aging protocol exceeds the constrained-East stationary-equilibrium baseline at every declared cell by at least the declared margin $3/2$ (achieved minimum ratio ≈ 1.6392 ; Table 1), and the audited idempotence-defect table exceeds its declared margin 2 (achieved ratio $289/121$). Grade: theorem (audited class); evidence type as declared in the artifact: deterministic float64 for the CD grid and its margin, exact rational for the idempotence-defect ratio and the lumpability controls. In the framework’s filing vocabulary the claim reads: the P5 channel of the standard thermodynamic packaging is not `active_projection`

as a closed packaging on this declared glassy scope; that is, packaging the present energy readout does not yield a closed description. Anchors: F-III T8, F-I D-IC-01/D-IC-02/T-CL-01, [75] Def. 3.1.

Table 1. GT1 closure deficit vs. constrained-equilibrium baseline (East $N=8$, L_{energy})

τ	CD_τ (glassy)	CD_τ (equilibrium)	ratio
1	0.033650	0.018875	1.7828
2	0.063730	0.036068	1.7669
4	0.113110	0.065575	1.7249
8	0.177212	0.108112	1.6392
margin	–	–	declared 3/2, achieved 1.6392

4.2 The control structure, including a corrected expectation

The controls deserve emphasis because one of them corrected our own design document. The design called for constrained-East equilibrium to serve as a zero-deficit control, on the intuition that an equilibrated chain has nothing left to remember. That intuition is *false*, and provably so. $CD_\tau = 0$ if and only if the energy readout is τ -lumpable, and whether it is depends on the kinetic constraint, not on stationarity: in East a spin can flip only when its left neighbor is up, so the number of spins free to move, and hence the rate at which the energy can change, depends on *where* the up spins sit, not merely on how many there are. So the energy readout of East is not closed even at equilibrium. The shipped certificate therefore uses two controls with different jobs (Fig. 2): the *unconstrained* model, whose energy readout is lumpable by construction, sits at exact rational $CD = 0$ (float residuals at 10^{-17} , exact lumpability verified symbolically), while constrained-East equilibrium is an honest *nonzero baseline*. The content of the certificate is that aging exceeds even that baseline by the declared margin. Equilibrium is not used as a zero control anywhere.

[GB2] Because the aging chain is driven by a temperature schedule, the closure deficit must be defined relative to the protocol rather than to a single stationary kernel. The machinery for that is the GB2 demo: finite protocol-relative CD computations (East versus unconstrained, $N = 4$, declared ϵ grid) with exactness reserved for the lumpability check and float64 windowed diagnostics labeled as such. GB2 is recognition-grade, a demonstration rather than a certificate, and the GT1 artifact carries a passing consistency gate for it: in the stationary limit the protocol-relative deficit reproduces the stationary deficit with residual exactly 0.

4.3 Scope and open control

The certificate quantifies over exactly its declared cell: East, $N = 8$, L_{energy} , the declared τ ladder and quench window. FA replication, the richer L_{panel} lens, and $N = 10$ are extensions that were not run, listed as such in the artifact envelope.

[NEG.gt1_b_parity_control_open] One intended control is open: the paper-[B] example-chain benchmark-parity check was not ported, and the GT1 artifact files it as an explicitly skipped control with its reason (machine-readable **status: skipped**). The certificate’s margins are internal to its own declared scope and do not route through [B]’s example chains; no [B]-parity is claimed until the control is actually run.

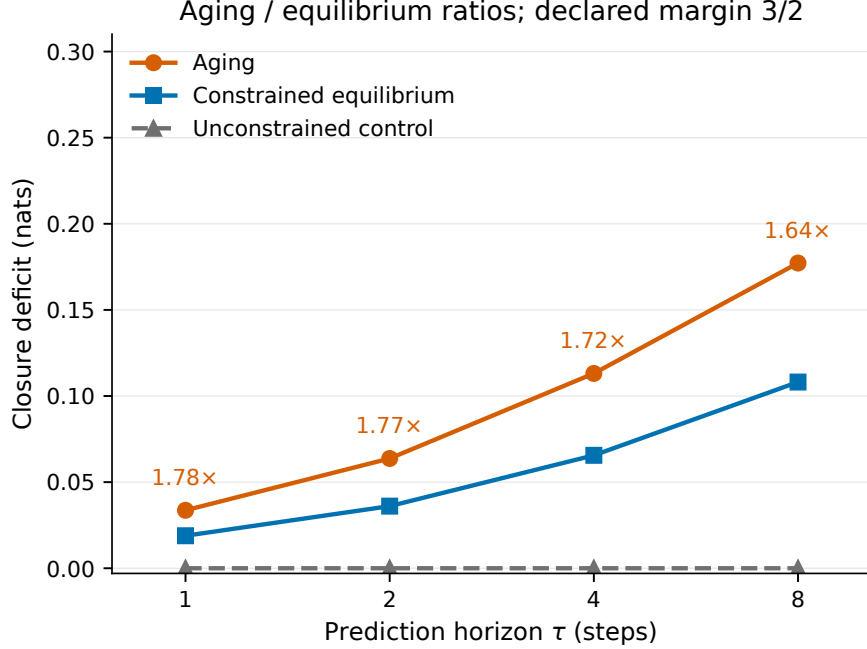


Figure 2. GT1 closure deficit (nats) versus horizon τ : aging protocol, constrained-equilibrium baseline (nonzero because the energy readout is not lumpable), and the unconstrained control at exact zero. Annotations give the aging/equilibrium ratio at each horizon against the declared margin $3/2$.

5 GT2 and GT5: The Kovacs Witness and What It Costs to Repair

5.1 The Kovacs protocol on the audited class

The Kovacs protocol [40] (equilibrate hot, quench cold, age for t_w , jump up to an intermediate temperature, watch a one-time observable relax) was located on the East substrate by an exact search, run before any certificate and filed as a pre-certificate control. At $N = 10$, $\epsilon : 7/10 \rightarrow 1/10 \rightarrow 2/5$ with $t_w = 40$, the mean energy after the second jump starts below its new equilibrium value $20/7 \approx 2.857$ (at ≈ 2.741), crosses it between steps 1 and 2 (≈ 2.803 and ≈ 2.859), and overshoots through a hump peaking at $t_{\text{peak}} = 35$ before returning (Fig. 3), all in exact rational arithmetic; $t_K = 1$ denotes the last step before the crossing. The raw trace never equals $20/7$ at an integer step; the preparation that does, a randomized stop between steps 1 and 2, is constructed in §5.2. The non-monotone excursion after the crossing is the classic signature that the present value does not determine the future; for a finite Markov chain in the linear-response regime, Prados and Brey showed it is generic whenever the observable’s relaxation is not a single exponential [55]. That East- and FA-type models show a Kovacs hump was established numerically by [16, 4]; what is new here is the exact-rational trace and the certificate built on it.

5.2 GT2: the witness certificate

In plain terms: we exhibit two preparation histories of the East chain with exactly the same present mean energy and provably different mean energies five steps later. The first history is the Kovacs run stopped at the crossing: since no single time step lands exactly on $20/7$, the run is stopped at step 1 with probability $\theta \approx 0.0359$ and at step 2 otherwise, with θ solved exactly so that the mixture’s mean energy is $20/7$ (a “randomized stop”, a declared move of the pipeline). The second history is the equilibrium distribution at $\epsilon = 2/5$, whose mean energy is $20/7$ by definition. Both are

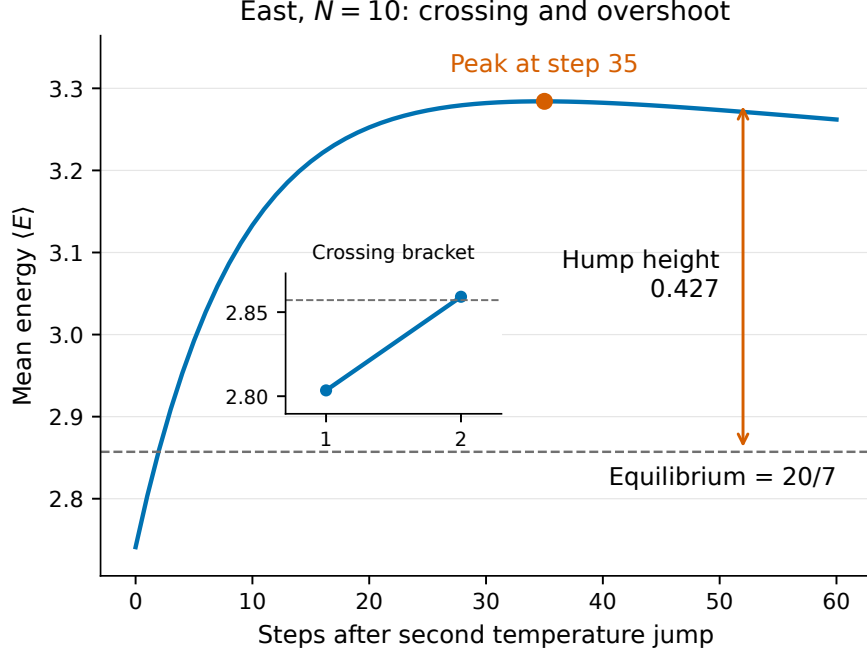


Figure 3. Kovacs hump on East $N = 10$ (exact-rational trace, floats only for plotting): the mean energy starts below its equilibrium value $20/7$, crosses it between steps 1 and 2 (inset), peaks at step 35, and returns toward equilibrium.

then held at $\epsilon = 2/5$ for five steps and their mean energies compared: the equilibrium history stays at $20/7 \approx 2.857$ while the Kovacs history has moved to ≈ 3.058 , a gap of ≈ 0.201 . Only the mean energy is matched at the start, not the full energy distribution; that is exactly what the energy readout, as a lens, sees. Precisely:

[GT2] On the declared scope (East, $N = 10$, `kovacs_moment` lens, declared ϵ grid and continuation catalog), one exact-rational split pair of preparation histories is exhibited: the two histories agree on the present lens value (mean energy $20/7$) and their futures differ, with an exact maximum absolute future gap that is a strictly positive rational number (≈ 0.2005 on the five-step hold; the exact fraction is in the artifact). By [74] Thm. 3.5, the existence of such a pair is equivalent to the comparison map $\pi : M \rightarrow Q$ being strictly refining, that is, to the predictive description being genuinely finer than the present-observable one. Grade: theorem (audited class).

The framework distinguishes three things a “different future” could mean, and we claim only the first: that the future depends on more than the present readout; *not* that the two histories took different routes to the same state (a P3 route mismatch); and *not* any directionality or irreversibility. In the framework’s filing vocabulary the first reading is **P4 staged-dependence**, filed on the standard witness template (F-II `prop:p1-witness`). No P6_drive audit was performed, so no arrow-of-time statement of any kind is made on GT2’s behalf.

[NEG.gt2_loop_certificate_not_built] The proposal also called for a companion certificate about thermal loops (a closed temperature cycle acting trivially on the present description Q but nontrivially on the predictive description M). It was not built: the split-pair witness stands alone, and, as §9 makes precise, the vendored formal core cannot even express a non-vacuous loop-asymmetry instance, so a computational loop certificate would first need new formal footing. Disclosed here and in the gap register; nothing downstream depends on it.

5.3 GT5: what it costs to buy the distinction back

In plain terms: adding a single scalar to the energy readout is enough to tell the two witness histories apart, and a fictive-temperature-like index is one of three declared choices that work. Precisely:

[GT5] On the declared scope (East, $N = 10$, `kovacs_moment` interface, the declared two-history class), the present lens collapses the witness pair into one class ($|Q| = 1$ on the pair) while the predictive quotient keeps two ($|M| = 2$), so the largest fiber of π has size `MaxFiber` = 2, meeting the nontriviality threshold. The buyback curve (Fig. 4) is published as a lift ledger per `F-II prop:forgetting-lifts`: each added memory coordinate is a recorded, selected, non-unique lift. At budget 0 the pair is unresolved; at budget 1, *each* of the three declared scalar coordinates already resolves it exactly (loss 0/1), so the shipped witness saturates at `saturation_budget` = 1. The three coordinates are expectations over the history’s microstate distribution μ of a per-microstate quantity: $T_f(\mu) = \mathbb{E}_\mu[i^*(E(n))]$, where $i^*(E)$ is the index of the grid value ϵ_i whose equilibrium mean energy is closest to E (an index into the declared grid, not a temperature, and not inferred from the history’s mean energy, so two histories with equal mean energy can have different T_f); $D(\mu)$, the expected number of adjacent up-up spin pairs; and $n_1(\mu)$, the expected occupation of the site next to the wall.

The translation to the glass literature is deliberately modest. The TNM fictive temperature [69, 49, 47, 33] is the motivating analogy for T_f : an energy-indexed effective temperature added as a declared step from the present description toward the predictive one. The implemented T_f is the grid-index construction above, not the TNM quantity itself. Modern treatments of the Kovacs effect likewise pair an effective temperature with a second structural variable [14], and the multiparameter models the Kovacs literature moved to [41] are why the general insufficiency question is left open here rather than answered on one pair. For *this* minimal two-history witness, one scalar suffices; the certificate demonstrates the repair *mechanism* and its ledger discipline. It is not a claim that a single fictive-temperature-like scalar is insufficient in general: such a claim would require a richer multi-instance construction that was not built here.

6 GT3 and GT4: Foreclosure and the Kill-Test Panel

6.1 GT3: the order-parameter search, foreclosed on its declared class

In plain terms: a static order parameter built from the energy readout is a function of the energy value. On $N = 8$ spins the energy takes 9 values, so there are exactly $2^9 = 512$ yes/no functions of it, and we can check them all. Every one of them agrees on a pair of histories that a finer readout of the same microstate, the occupation of the site next to the wall, tells apart. Precisely:

[GT3] On the declared scope (East, $N = 8$, $\epsilon = 1/2$), the definability bound of F-III T14 makes the static-order-parameter search space finite: a fixed lens with image size 9 defines exactly $2^9 = 512$ predicates. The sweep checks all 512: every one agrees on the exhibited witness pair, while a distinguishing readout based on n_1 (the occupation of the site next to the wall) separates it, with exact gap 169/1536 on the separating continuation (Table 2). The witness pair is a standalone reflection pair constructed for this sweep, deliberately *not* chained to the GT2 certificate, so the two results fail independently. This is a constructive foreclosure: the distinguishing readout is exhibited and proven absent from the set of energy-definable predicates, not inferred from counting. Grade: theorem (audited class).

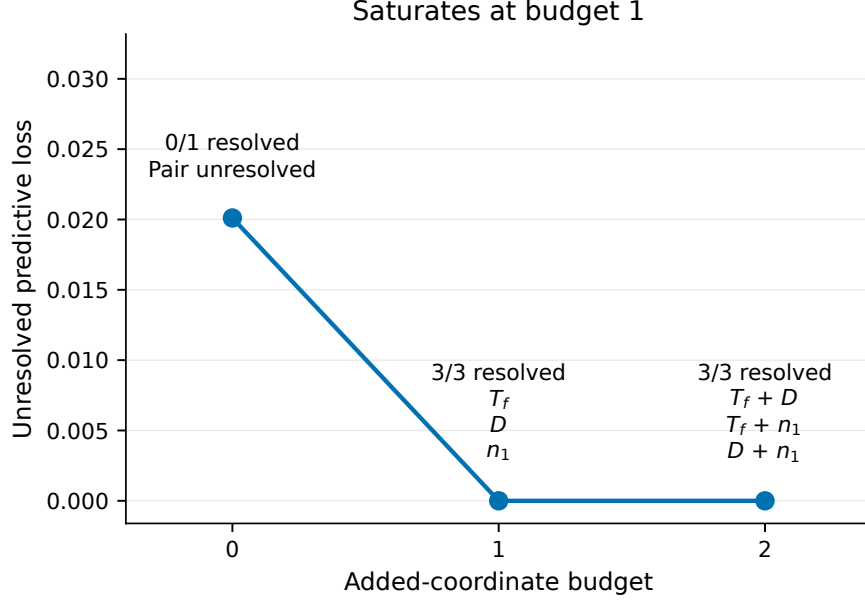


Figure 4. GT5 buyback curve: unresolved predictive loss versus added-coordinate budget, annotated with resolved/attempted counts and the resolving coordinates. The loss is the budget envelope of the two-point squared prediction error $(a - b)^2/2$, where a and b are the two histories’ five-step future mean energies, incurred only while the pair is left in one class (≈ 0.0201 at budget 0, i.e. half the squared witness gap); a group that separates the pair has loss 0. The shipped two-history witness saturates at budget 1.

Table 2. GT3 foreclosure sweep summary (East N=8, $\epsilon = 1/2$)

field	value
N	8
epsilon	1/2
lens	L_energy
total predicates	512
agreeing predicates	512
lens image size	9
all agree	True
separating event	n_1
witness provenance	standalone W_reflection witness constructed for this sweep; not chained to the GT2.kovacs_moment certificate (a separate witness family per design/specs/04_gt2_witnesses.md §4).
T-FOR-02 statement	the distinguishing readout (n_1-based) is not in Def(l_energy); constructive witness exhibited, not generic counting

The field-facing sentence, “there is no static order parameter for glass,” is licensed here *only* as the declared-class shadow of this theorem (F-II item E2), with the class, the catalog, and the suppressed content named in the same breath: no predicate definable from the declared energy lens on the declared class separates what memory separates. Reading it as a statement about other instruments, other lens classes (in particular the amorphous order of random first-order transition

theory [39, 42], the configurational-entropy line descending from [1], or the local-structure measures surveyed in [61]), or the continuum is not licensed, and this paper does not do so. What the theorem does license, via the framework’s no-free-distinction law (F-IV F12), is narrower and exact: any readout that *does* separate this pair must carry declared information beyond the energy readout. Here that information is a present structural coordinate of the microstate (n_1 , under the identity continuation), not a record of the history; the separation shows that the energy-based description is incomplete, and GT5 separately shows that one added coordinate, structural or history-based, suffices to repair it on the Kovacs pair.

6.2 GT4: the kill-test panel

A memory signature can be faked. GT4 builds five schematic fixtures for the ways it can be faked, each a small package of two or three microstates at $N = 4$ with declared readouts and continuations, and checks that the same deterministic classifier that accepts the Kovacs candidate (a reflection pair at $N = 8$, distinguishable by the full spin panel but not by its reflection-invariant part) rejects all five. The fixtures are regime schemata, not physical mechanisms: the “protocol trap” pair carries a residue of its schedule in the readout, and its “honest” twin replaces the identity continuation by a constant reset so the residue is no longer future-visible; the “flattenable” pair does the same for an incomplete continuation catalog; the “explicitly latent” pair differs in a bit the present readout does not see; the “dissipative” fixture merges the two histories at a later interface; and the “flat” control is two copies of the $N = 4$ equilibrium distribution, which by construction has nothing to separate. Precisely:

[GT4] Over the declared 9-member thermal benchmark family, the deterministic triage classification identifies `kovacs_wheel`, the Kovacs memory candidate, as the sole `coherent_candidate`, while the five declared non-coherent regime controls (`artifact_trap`, `dissipative`, `explicit_latent`, `flat`, `flattenable`) are correctly killed or cleared (Table 3). Protocol-trap discipline is applied: raw and cleared classifications are both recorded, so the table shows each artifact being caught (for instance, the naive protocol-trap fixture classifies as `artifact_trap` raw and its constant-reset twin as `flat`). Grade: theorem (audited class), over the declared finite family only, not a shell-general theorem.

Table 3. GT4 regime triage over the declared benchmark family

regime	raw classification	cleared classification
coherent candidate	<code>kovacs_wheel</code>	—
<code>artifact_trap</code>	<code>protocol_trap_naive</code>	<code>protocol_trap_honest</code>
<code>dissipative</code>	<code>dissipative_memory@mid</code>	<code>dissipative_memory@end</code>
<code>explicit_latent</code>	<code>latent_memory_base</code>	<code>latent_memory_refined</code>
<code>flat</code>	<code>flat_control</code>	(pure control, no repair)
<code>flattenable</code>	<code>flattenable_raw</code>	<code>flattenable_completed</code>

6.3 GB1: the protocol-trap caveat, carried openly

The “protocol trap” is a warning about arrows of time, not about memory: a system whose dynamics switch between several kernels under a clock can show a spurious time asymmetry if the clock is left outside the state. The framework’s published theorem on this (F-I T-AOT-02) says that when the clock is reversible and every phase kernel is reversible with respect to a *common* stationary measure, the autonomous lift of the system has zero entropy production and zero path-reversal asymmetry at

every horizon, so any observed arrow must come from an external schedule or a stroboscopic fit. Kovacs kernels K_ϵ are reversible with respect to different Gibbs measures for each ϵ , so the theorem does not apply as published, and the temperature switches contribute a nonzero boundary term.

[GB1] GB1, the temperature-protocol version in the style of the protocol-trap microarticle [78], ships as a recognition-grade mathematical demonstration: an exact identity for that boundary term (a pseudo-entropy-production identity) plus a finite exact numerical check, with the general inequality filed as an explicitly open obligation in the artifact payload. Its role is to keep the arrow-of-time audit separate from the memory classification: GT4’s triage computation is deterministic and exact, but the clock-audit *discipline* it applies (raw versus cleared classification) rests, for temperature protocols, partly on this demonstration rather than on a closed theorem. This is the framework’s conditional-reporting rule working as designed.

7 GT6: The Memory Layer Strictly Extends the Observable Layer

7.1 Claim and verdict basis

In plain terms: describing an East glass by the future-equivalence class of its preparation history carries information that the declared energy-based description cannot, and the framework’s checklist for certifying such a statement (a gate table with pass/fail rows, plus ablations) is completed for the audited shell, the frozen eligibility conditions defined in §2. Precisely:

[GT6] For the audited East glass shell ($N = 8$, declared ϵ grid and protocol windows), the promotion filing of F-III §10 (bridge record, gate table, knockout panel, nonclaim register, checker) is complete, and the gate-table verdict is **strict**: the predictive-structure layer strictly extends the current-observable layer on this shell. Grade: certificate (shell-local, the standard of paper [C] [76]): valid on the frozen shell, with no broader-class transfer claimed.

Because an earlier draft of this project’s own design documents overstated this result, the basis of the verdict is quoted verbatim from the audit artifact rather than paraphrased:

“verdict: strict is established via F-III T19/T20 (non-factorization from GT2/GT3 witnesses). Paper [C]’s thm:strict-extension (the fuller completion-dynamics route requiring material $P4 \leftarrow P5$ forcing) is NOT invoked: `H4_forcing.material_forcing_count = 0` for this shell, filed as an honest negative sub-result, not as a satisfied hypothesis.”

That is: the strictness verdict rests on *non-factorization*. GT2 exhibits histories that the energy-based description merges and the predictive one separates, and GT3 shows that no energy-definable predicate can undo the merge, so the predictive description cannot be a function of the energy-based one; the framework’s theorems T19/T20 turn that into the verdict. The verdict does *not* rest on paper [C]’s stronger completion-dynamics theorem, whose “material forcing” hypothesis this shell simply does not satisfy. (Material forcing means that the catalog of protocols contains *conditioned* schedules, in which the next temperature step depends on the measured panel, as in real glass processing, and that some distinctions between histories exist only because of them. The audited catalog produced none.)

[NEG.material_forcing_absent] The forcing check itself is a result: `material_forcing_count = 0` on the audited shell. Absence of material forcing here is not evidence about other shells or protocols; it marks the exact boundary of which [C] machinery this instantiation exercises.

7.2 Gate table and knockout panel

The gate table (Table 4) lists the framework’s required checks for a strict-extension verdict and the evidence for each; two rows are marked not required because the claim is about enrichment of the description, not about a closed macroscopic dynamics.

Table 4. GT6 gate table (audited glass shell)

gate	required	status	evidence
G_suff	True	pass	audit.json filing assembled
G_desc	False	not_required	glass bridge claims strict object-map enrichment, not closed macro dynamics
G_stab	True	pass	P3-04 saturated strata under E_tau,f
G_ctrl	True	pass	P3-01 knockout panel controls
G_nosmuggle	True	pass	six no-smuggling violation kinds checked
G_vis	True	pass	visible citations and nonclaims filed
G_audit	True	pass	checker validates hashes and verdict
G_strict	True	strict_pass	GT2/GT3 constructive split witnesses
G_logglob	False	not_required	shell-bound single bridge only; no stacked or glued promotions claimed

The knockout panel (Table 5) is an ablation study: each of the six primitives P1–P6 that the framework says a closed description can depend on is disabled in turn, and the effect on the result is recorded, in two distinct modes that the artifact’s own nonclaim keeps separate. For P1–P3 the knockouts are *numeric*: disabling the primitive (identity run kernels; the unconstrained-lumpable replacement; constant- ϵ replacement) degrades a declared numeric diagnostic past its declared threshold. For P4–P6 the knockouts are *capability-loss* rows: removing the primitive makes the relevant downstream diagnostic unavailable or undefined rather than numerically degraded. There is no numeric threshold to cross in those rows, and none is claimed.

Table 5. GT6 knockout panel

primitive	kind	degraded	threshold
P1	numeric	True	identity trace variation must be exactly 0
P2	numeric	True	unconstrained control must be exactly lumpable under L_energy
P3	numeric	True	constant-epsilon replacement must fail the hump detector
P4	capability_loss	True	–
P5	capability_loss	True	–
P6	capability_loss	True	–

Supporting diagnostics are filed at their true, lower grades: the Fekete pressure-closure table is a float64 diagnostic (not an exact-rational certificate; sparse $N = 10$ rows skipped by declared threshold and reported as such), and the spectral-gap table (§3) calibrates the shell windows only.

The gate row H5 (the macro-admissibility obstruction) cites the GT1 theorem-grade certificate by content hash; paper [C] itself glosses this obligation as failure of admissible aggregation or lumpability at the macro level, which is precisely what GT1 certifies.

7.3 What strictness does not mean

The artifact’s nonclaim register (printed in full in §11) is part of the result. In particular: strictness is shell-local (no shell-general strict-bridge theorem, no broader-class transfer); strictness does not imply that the memory layer has an autonomous closed dynamics of its own, nor that anything drives it (no P6_drive audit was performed); the pressure gap is neither a closure deficit nor the strictness witness; and the promotion is a single bridge $T_0 \rightarrow T_1$. The framework has no theorem for stacking such bridges, and none is used.

8 GT7: Aging Constants Are Run-Only: Freeze, Transfer, and the Scoreboard

8.1 The protocol

GT7 asks whether the numbers that characterize aging on East (Kovacs crossing time, peak time, hump height, and three others) carry over to a different model without any fitting. The answer was obtained the way a pre-registered experiment is [50, 48]: the readouts were fixed, the East reference was computed, predictions for the target models were committed and hashed, and only then were the targets run and scored by a script that had itself been frozen. Precisely:

[GT7] GT7’s claim is a discipline plus its unedited outcome. The freeze-and-transfer protocol ran in five stages, each frozen by sha256 content hash before the next began:

- **F1, readouts freeze.** Six declared readouts (Kovacs t_K , t_{peak} , hump height, memory-dimension profile, deficit-decay slope, cross-validation checks) frozen in a registry with content hash and freeze note.
- **F2, East reference.** The reference surface on East: 56 exact hump-surface cells at $N = 10$, an $N = 16$ Monte Carlo spot check, the closure-deficit ladder (Fig. 5) whose declared secant slope is the frozen deficit-decay readout, and memory-dimension profiles. Run-only readouts; no functional form fitted anywhere.
- **F3, pre-registration.** 338 predictions committed before any target-family run: 42 interval predictions, 2 order-relation predictions, and 294 cells explicitly declared `no_prediction`. The predictions registry pins the frozen readouts hash, the East reference hash, and the hash of the scoring script itself.
- **F4, target runs.** FA and trap families run blind against the committed predictions.
- **F5, mechanical scoring.** The frozen scorer (hash-pinned at F3, verified unchanged) compares predictions to observations with no post-hoc widening or reinterpretation.

8.2 The scoreboard, unedited

[NEG.gt7_transfer_failures] Of the 44 committed predictions, 22 were scored against observed target cells: **8 passed and 14 failed**; the other 22 targeted cells that produced no observation. All 338 verdicts are filed unedited (Fig. 6). The failure pattern is itself a filed, tested fact:

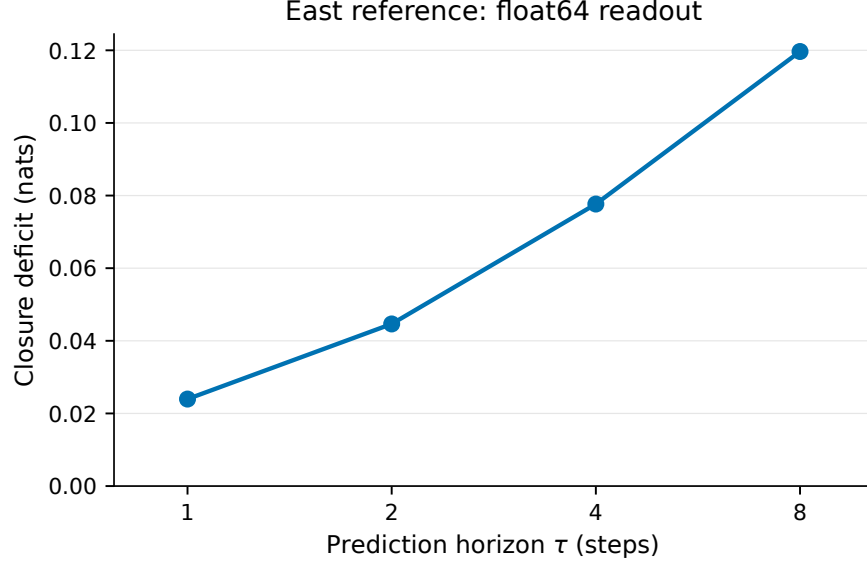


Figure 5. East-reference closure deficit (nats) across the declared τ ladder (run-only float64 readout; part of the frozen F2 reference). The frozen “deficit-decay” readout is the declared secant slope between two ladder points, not a fit; the deficit itself grows with horizon.

- All 14 failures are FA-family interval predictions (5 t_K , 4 t_{peak} , 5 hump-height), and every one is *one-sided high*: the observed FA value exceeds the predicted upper bound. FA humps are systematically later and larger than the East-anchored intervals allowed.
- Failures cluster at the deeper intermediate temperature: 12 of 14 failing cells have $\epsilon_{\text{mid}} = 2/5$, where every scored cell failed. At $\epsilon_{\text{mid}} = 1/2$, 7 of 9 scored interval cells passed (plus the passing order-relation prediction), and the two misses there are borderline (hump height 0.2033 against a bound of 0.2008; t_K of 5 against $[1, 3]$).
- All 22 unobserved cells are trap-family: the trap target’s exact hump search returned `found = false` in all 56 surface cells, with the uniform miss reason “does not start below equilibrium.” This is the dynamical corollary of the ϵ -invariance lemma of §3: since the declared trap family’s equilibrium does not move with ϵ , a temperature jump never leaves it below its new equilibrium, and the Kovacs protocol cannot produce a hump there at all.

8.3 Reading the failures

The failures are first-class results of the pre-registration protocol: it is designed to be falsifiable, and it was. Three readings are compatible with the data, and the protocol forbids choosing among them after the fact: (a) East-to-FA transfer of run-only constants degrades with quench depth, in the direction of FA responding slower and larger; (b) the declared rule for scaling East intervals to FA, rather than the readouts themselves, is what failed; (c) the trap column is not a partial failure but a structural mismatch between the protocol and the declared trap family, predicted in hindsight by the ϵ -invariance lemma. Under any reading, the run-only thesis stands as filed: the readouts are readouts of runs, frozen, transferred, and scored, not fitted. (GT3’s non-definability result concerns its own witness pair and the energy lens; it is not extended here to the GT7 readouts, for which no such witness was constructed.) Score counts are not a fitted scaling law, and no post-hoc reinterpretation of failed intervals is permitted or performed.

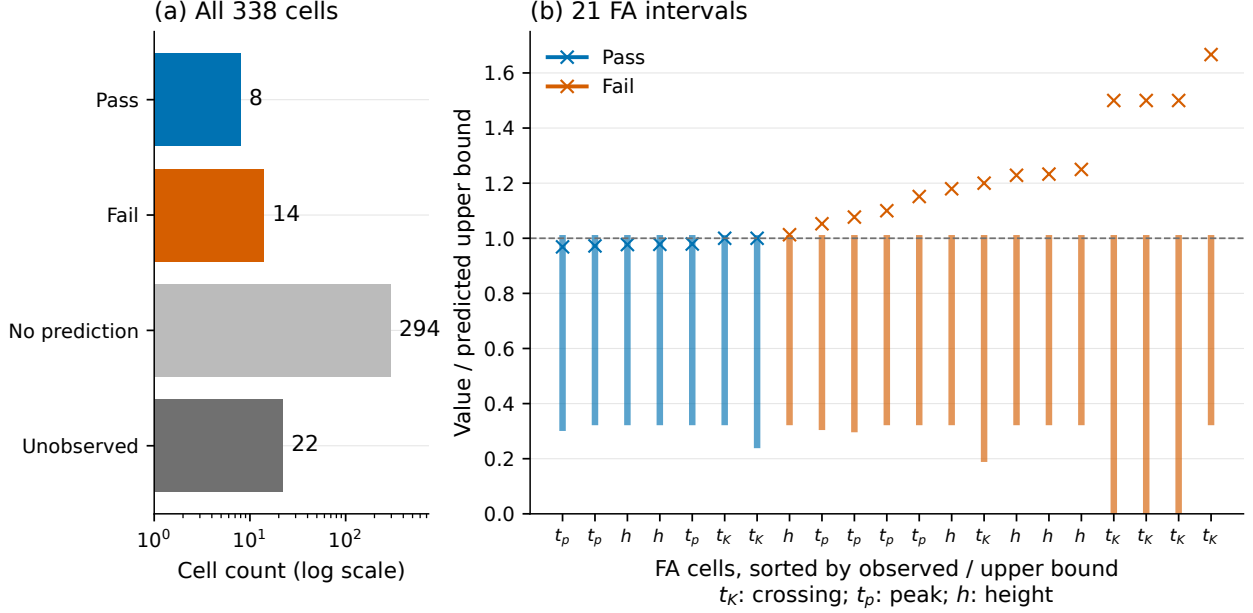


Figure 6. Left: verdict counts over all 338 cells (log scale). Right: the 21 scored FA interval cells (the 22nd scored FA prediction is the order relation, not plotted); each interval and its observed value are divided by the interval’s upper bound, and cells are sorted by that ratio, so every failure lies above the dashed line at 1.

9 GB9: The Lean-Checked Core and the Scope-Boundary Finding

9.1 What is machine-checked

Lean [25] is a proof assistant: a theorem stated in it is accepted only when a machine has checked every step, and a development “compiles sorry-free” when no step has been left as an unproved placeholder. We use it for the abstract theorems the certificates rely on (for example, that a witness pair exists if and only if the predictive description is strictly finer than the present one), so that those steps do not depend on a reader trusting the prose. Precisely:

[GB9] The vendored **HolonomyMemory** core (paper [A]’s Lean development [74]) plus this project’s **GlassWitness** extension compile sorry-free on the pinned toolchain (`leanprover/lean4:4.29.0`), with per-theorem axiom audits. Per the framework’s fidelity legend, every row of the canonical 17-row fidelity table (Table 6) carries its true label: 11 rows *verified* (including the three generic collapse theorems below and [A]’s witness/strict-refinement equivalence), 3 *narrowed/parametric* (core structure fields), 1 *typed mirror* (the glass instance), 1 *schema projection*, and 1 *computational evidence only* (the numeric Kovacs witness, which is deliberately not encoded; see below). No schema or trivial declaration is described as a verified proof anywhere in this paper, and closure-deficit computations are deterministic computational evidence, not Lean theorems.

Table 6. GB9 Lean fidelity table (all rows, printed in full per the Phase 5 gate; rows with no Lean encoding are labeled by the claim they document instead of a Lean identifier)

Lean name / documented claim	fidelity
futurePredictiveEquiv_implies_currentEventEquiv	verified
predictiveToCurrent_commutates_with_transport	verified
predictiveQuotient_futureSufficient	verified
futureSufficient_stateEq_implies_futurePredictiveEquiv	verified

Lean name / documented claim	fidelity
<code>futureSufficient_factorsThroughPredictiveQuotient_onReachable</code>	verified
<code>futureSufficient_factorization_unique_onReachable</code>	verified
<code>strictRefinement_iff_nonempty_predictiveWitness</code>	verified
<code>loopAsymmetry_exhibits_movedPredictive_fixedCurrent</code>	verified
<code>current_implies_future_GENERIC</code>	verified
<code>predictiveWitness_impossible_GENERIC</code>	verified
<code>strictRefinement_impossible_GENERIC</code>	verified
<code>push_id</code>	narrowed / parametric
<code>observe_push</code>	narrowed / parametric
<code>push_compose</code>	narrowed / parametric
<code>glassInstance</code>	typed mirror
<code>glassPredictiveQuotient_futureSufficient</code>	schema (projection)
Real Kovacs exact-rational numeric split-pair witness	computational evidence only

9.2 The collapse finding

We intended to encode the numeric Kovacs witness of §5 in the core as well. The attempt produced a theorem instead: in this core, no such witness can exist. Precisely:

[FINDING.lean_collapse] For *every* instance of `RouteTransportCore`, the `observe_push` coherence field, combined with `CurrentEventEquiv`’s universal quantification over events, forces current equivalence to imply future-predictive equivalence (`current_implies_future_GENERIC`); consequently non-vacuous `PredictiveWitness`, `StrictRefinement`, and `LoopAsymmetry` instances cannot be constructed in this core at all (`predictiveWitness_impossible_GENERIC` and its strict-refinement companion; machine-checked, sorry-free). The mechanism: the core’s naturality axiom lets every future event be pulled back to a present event, so “agreement on all present events” silently includes agreement on all pulled-back futures, and there is nothing left for a witness to separate.

This is a genuine result about the formal core’s expressiveness, strictly stronger than paper [A]’s own declared boundary, and it has a familiar shape. It is the proof pattern of strong lumpability for Markov chains [38, 15], where a local one-step coherence condition propagates an equivalence to all future times; and it has the form of biextensional collapse in the Chu-space sense [5, 56, 52], with `pullEvent` playing a role analogous to the Chu transform’s naturality condition. We offer the Chu-space reading as an analogy; no formal correspondence is claimed or proved here. The instructive contrast is the causal-state construction of computational mechanics [63], where agreement on the present generally does *not* imply agreement on the future: the collapse here is forced by this core’s axiom, not a generic feature of prediction.

9.3 The consequence, stated as understanding rather than apology

[NEG.lean_instance_structural] The finding explains exactly which certificates can live in Lean and which cannot. The abstract theorems (witness if and only if strict refinement; factorization through M and its uniqueness on the reachable image) are verified generically and apply to any instantiation. But a non-vacuous *numeric* witness instance is unbuildable in this core, so `glassInstance` is filed as what it is: a typed, index-based structural mirror, with the real Kovacs split pair remaining exact-rational computational evidence in its own artifact (§5). The Python certificates avoid the collapse because they quantify over a finite *declared* catalog of futures rather than over all pullback-closed events. A formal core admitting numeric glass witnesses would need weaker coherence axioms;

that is future formalization work, now with a machine-checked specification of exactly what must change.

10 GB10: Empirical Bridge Records

Nothing in the preceding sections is a measurement. The connection to the laboratory is made through three *bridge records*, drawn from the memory-effect phenomenology reviewed in [37], each of which states what a real instrument sees, what it suppresses, and under what threshold a measured signature would count as the laboratory counterpart of one of our certificates. Precisely:

[GB10] Three empirical bridge records connect the formal complex to laboratory phenomena, each filed with the complete five-field schema of F-II `prop:empirical-bridge` and validated by the bridge checker: (i) **Kovacs PVAc volume recovery** [40, 67, 65, 7], dilatometric specific-volume measurement under the temperature-jump/age/second-jump protocol, with the level map tying the measured hump to the GT2 witness artifact by content hash; (ii) **spin-glass memory and rejuvenation** [36, 13, 79]; (iii) **colloidal-glass aging** [80, 22, 21, 53]. Each record carries: substrate and observable; instrument with a per-instrument visibility record (what the apparatus sees and what it suppresses; dilatometry, for instance, sees one-time specific volume and hump shape while suppressing the chain-conformation microstate); a level map with formal-artifact hashes; a threshold-witness relation stating when a measured signature *counts* as a laboratory-grade witness; and suppressed-structure nonclaims.

The bridges are the paper’s only channel to the laboratory, and they are deliberately one-way: a measured Kovacs signature that meets the threshold is a laboratory-grade witness via the bridge, and nothing more is inferred from it. The laboratory counterpart of GT5 inherits GT5’s scope (§5): what is licensed today is that measured Kovacs memory shows the present description to be incomplete, not that a single fictive-temperature-like scalar is insufficient. A laboratory replication targeting these records would need the declared protocol grid as an instrument-realizable schedule, a declared one-time observable panel, and a matched-pair preparation (two histories tuned to equal present panels). A measured divergence of futures would then file as a laboratory-grade split pair through record (i)’s threshold-witness relation.

Nonclaim: No empirical-realization claim anywhere: laboratory data is never presented as a realization of the formal description. Visibility records are per instrument, with no cross-instrument merge into an instrument-free statement (F-II has no cross-instrument comparison theorem, and none is used).

11 Discussion

11.1 What is established, at which grade

Theorem-grade, each on its exactly declared finite class: the energy readout is not closed, with a certified margin over the equilibrium baseline (GT1); the Kovacs split-pair witness (GT2); foreclosure of all 512 energy-definable predicates on the declared class, with a finer present-state readout as the exhibited separator (GT3); the kill-test panel with five look-alike regimes correctly rejected (GT4); `MaxFiber = 2` with a saturating buyback ledger (GT5); and, from the formalization effort itself, the collapse theorem for `RouteTransportCore` (machine-checked). Certificate-grade, shell-local: the GT6 strict extension via non-factorization. Run-only: the GT7 protocol and its 8/14/294/22 scoreboard. Recognition-grade: the GB1/GB2 demonstrations, the Lean fidelity filing, and the three empirical bridges. Nothing in this paper is graded higher than its artifact.

11.2 What is open

Seven gaps are disclosed, each with a register entry stating what closing it would take (`paper/gap_register.md`); none is load-bearing for any claim above, because every claim’s wording already excludes it.

[NEG.gb5_repair_sweep_not_built] The largest is GB5: no exhaustive sweep over a frozen class of candidate repairs exists, so no repair-impossibility certificate of any grade is claimed. “No admissible repair exists” is never asserted, and GT5’s buyback saturation is explicitly not a substitute. The others: the GT2 loop certificate (unbuilt, and now known to need new formal footing); GT1’s [B]-parity control (skipped, filed); GT1’s breadth (one declared (model, N , lens) cell, which is the declared-class discipline working as designed, but honestly a single cell); the GT7 failure interpretation (three readings left open by design); material forcing (absent on this shell, so [C]’s fuller route is unexercised); and the numeric-witness Lean encoding (proved impossible in this core). Natural next experiments follow directly: a second pre-registered prediction round with re-derived intervals; an Arrhenius-form trap family (per-trap rather than common temperature dependence) so the Kovacs protocol is non-degenerate there; the FA, L_{panel} , and $N=10$ extensions of GT1; and a weakened formal core admitting numeric witnesses.

One ceiling is declared rather than open, per the framework’s recognition-open caveat: if glass aging carries a finite conservation or balance law over the declared records that supplies none of the six witness types, the six-role decomposition underlying this paper’s filings is conditional on the recognition hypothesis that no such feature exists. No such candidate was identified on the audited class; any future one files at recognition grade, separately.

11.3 Anti-reductionism, stated plainly

This paper’s certificates live at a boundary between two levels of description, the present-observable level and the predictive level, and the discipline that produced them cuts both ways. A passing certificate suite never proves emergence, and none of the information-theoretic emergence criteria in the literature [34, 60] is invoked: GT3’s foreclosure says that the declared lens class provably cannot express the distinction that memory carries, which is the method working as intended, not a metaphysical discovery about glass, and not a reduction of either level to the other. Symmetrically, nothing here licenses reading the strictness of the memory level (GT6) as autonomy, drive, or an arrow of time: no P6_drive audit was performed, and a positive closure deficit is neither novelty nor irreversibility on its own. The paper makes no claim about the existence or non-existence of a thermodynamic ideal-glass transition, takes no side in the field’s thermodynamic-versus-dynamical debate [11], and makes no continuum-limit claims; every quantifier in every claim ranges over a declared finite class, and the empirical legs are bridge claims only.

11.4 Full nonclaims ledger

Per the project’s own publication gate, the complete nonclaims ledger, every nonclaim of every claim row, unabridged, is printed here rather than summarized.

Table 7. Full nonclaims ledger (printed in full per the Phase 5 gate)

claim	nonclaim
GT1	GT1 certificate is scoped to East $N=8$ and L_{energy} only; FA and L_{panel} are undone extensions.

claim	nonclaim
GT1	The tau ladder is the declared finite ladder [1, 2, 4, 8], not the full combinatorial grid.
GT1	Constrained East equilibrium under L_energy is a nonzero stationary CD baseline; only the unconstrained lumpable control is exact-zero.
GT1	[B] benchmark-parity control is filed as skipped in this artifact because the paper-specific example chains were not ported in this remediation packet.
GT2	filed as P4 staged dependence, not a directionality or irreversibility claim
GT2	Phase-0 Kovacs search output is a pre-certificate diagnostic, not a GT witness certificate.
GT2	The selected result freezes a later config but does not by itself prove a theorem-grade claim.
GT3	finite N=8 East foreclosure over Def(l_energy), not a continuum or all-lens claim
GT4	GT4 triage is over the declared finite benchmark family, not a shell-general theorem.
GT4	The artifact promotes the Phase-1 benchmark-suite data; it does not change the underlying classification logic.
GT4	GB1 protocol-trap caveat remains a theorem-wrapper limitation; this table files the deterministic triage computation.
GT4	Phase 1 benchmark parity is a control suite, not a standalone GT4 triage certificate.
GT4	Dedicated GT4 regime-table promotion, if present, is filed in a separate artifact.
GT5	buyback saturation or nonsaturation is not a GB5 repair-impossibility certificate; GB5 requires a separate exhaustive repair-class sweep
GT5	the shipped Kovacs buyback curve is a minimal two-history witness with saturation_budget=1; it demonstrates buyback mechanics, not general single-scalar or TNM insufficiency
GT6	GT6 is scoped to the audited glass shell, not to all East or glassy regimes.
GT6	No shell-general strict-bridge theorem is claimed.
GT6	No broader external-class theorem is claimed beyond the frozen finite class.
GT6	Strictness does not imply autonomous macro closure.
GT6	Strictness does not imply drive; P6_drive requires the separate GB1 audit discipline.
GT6	The pressure gap is not a KL closure deficit unless an explicit bridge is supplied.

claim	nonclaim
GT6	The pressure gap is not the strictness witness; GT2/GT3 provide the strictness witnesses.
GT6	The gap functional is a conditional disintegration on strata, not stratumwise root separation.
GT6	This is a model realization, not a universal strict-extension theory.
GT6	T1 is a packaged completion layer, not repeated application of one fixed idempotent completion.
GT6	Single bridge only; no stacked or glued promotions are claimed.
GT6	strict-extension inputs are shell-local and do not claim continuum or all-protocol closure
GT6	H5 and non-factorization are cited from prior artifacts, not recomputed here
GT6	Knockout rows are shell-local diagnostics, not shell-general strict-extension proofs.
GT6	P4-P6 rows are capability-loss degradations, not numeric re-simulation degradations.
GT6	Pressure closure is a float64 Fekete diagnostic, not an exact-rational certificate.
GT6	N=10 dense pressure rows may be skipped by declared threshold and are reported as such.
GT6	Spectral gaps are float64 diagnostics and not theorem-graded exact-rational certificates.
GT6	Mixing-time bounds are used for shell-window calibration only.
GT7	run-only readouts, not a fitted scaling law
GT7	East reference only; no transfer claim without a scored FA/trap target run and a pre-registered prediction.
GT7	FA target run only; scoring is deferred to the frozen transfer scorer.
GT7	memdim_profile and deficit_decay transfer are out of scope for this target artifact.
GT7	Trap target run only; scoring is deferred to the frozen transfer scorer.
GT7	Trap uses a declared depth observable, not East/FA spin-count energy.
GT7	transfer scoring is mechanical comparison against pre-registered predictions
GT7	score counts are not a fitted scaling law
GB1	GB1 demo records a pseudo-EPR identity and finite numerical check, not a closed general theorem.
GB1	The open obligation remains filed in the payload and is not discharged by this artifact.
GB2	GB2 demo records finite protocol-CD computations, not a GT claim certificate.

claim	nonclaim
GB2	Windowed and marginalized CD values are float64 diagnostics; exactness is reserved for the lumpability check.
GB9	The real Kovacs numeric split-pair witness is not encoded in Lean.
GB9	No non-vacuous PredictiveWitness, StrictRefinement, or LoopAsymmetry instance is claimed.
GB9	The GlassWitness instance is structural and indexed, not a real numeric package export.
GB10	No empirical-realization claim anywhere; bridges are admissibility filings only.
GB10	Visibility records are per instrument; no cross-instrument merge into an instrument-free statement.
FINDING.trap_epsilon_invariance	Not a novelty claim: the multiplicative-scalar cancellation is standard trap-model structure once stated; the contribution is stating and testing it for the declared family.
FINDING.trap_epsilon_invariance	Says nothing about trap dynamics or aging trajectories; only the stationary distribution is epsilon-invariant.
FINDING.lean_collapse	Not a claim that Kovacs memory is unformalizable; only that THIS core's coherence axioms foreclose non-vacuous witness instances.
FINDING.lean_collapse	Does not weaken the exact-rational computational witnesses; it explains why they are filed as computational evidence rather than Lean theorems.
NEG.material_forcing_absent	Absence of material forcing in this shell is not evidence about other shells or protocols.
NEG.gt7_transfer_failures	Failure counts are not evidence against the run-only thesis; they are the run-only thesis operating as designed. No post-hoc reinterpretation of failed intervals is permitted.
NEG.gb5_repair_sweep_not_built	'No admissible repair exists' is never asserted at any grade.
NEG.gt2_loop_certificate_not_built	No loop-asymmetry or P3 route-mismatch claim is made anywhere on GT2's behalf.
NEG.gt1_b_parity_control_open	No [B]-parity is claimed for GT1 until the control is actually run.
NEG.lean_instance_structural	The Lean track certifies the abstract core's theorems, not glass numerics.

12 Conclusion

On the exact-finite class built here, the observables one usually monitors for a glass do not form a closed description, and that failure is now carried by certificates rather than intuitions: a closure deficit exceeding its equilibrium baseline by a certified margin; an exact Kovacs pair of histories with

the same present and different futures; a constructive foreclosure of every predicate the declared lens can define; a kill-test panel that rejects its own constructed look-alikes; a priced repair of the lost distinction; a shell-local strict-extension verdict resting on exactly the theorems it needs and no more; and a pre-registered transfer scoreboard filed with its failures intact. Around the results sits the part we consider most transferable: a claims ledger tying every claim to hash-pinned artifact evidence, the bindings machine-checked at build time and the wording fidelity human-audited and independently reviewed, together with a nonclaims ledger printed in full.

The field kept asking “what is the order parameter of glass?” On the class where we can afford certificates, the certified answer is that the question is asked at the wrong level of description: the declared energy readout cannot express the distinction that the Kovacs pair’s futures carry, and the Kovacs effect has been the witness pair sitting in the literature since 1963. Whether that answer survives at larger sizes, richer observable panels, and ultimately in the laboratory is exactly what the disclosed gap register and the bridge records are built to test next.

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